Revisiting the Zig-zag graph product & USTCON and some of their applications to the complexity of space bounded computation

T. C. Vijayaraghavan[*] and **V. Arvind**[†]

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Abstract

In the year 2008, in his breakthrough paper [Rei08], Omer Reingold has described an $O(\log n)$-space algorithm which determines if there exists an undirected path connecting any two vertices s and t in an input undirected graph G. The main component of the algorithm of Omer Reingold is the Zig-zag graph product [RVW02, Rei08] using which it is possible to transform any undirected graph into an expander in $O(\log N)$ space, where N is the number of vertices in the input undirected graph G. In this paper, we examine the zig-zag graph product, discuss its properties and explain the $O(\log N)$ space algorithm of Omer Reingold [Rei08] along with some applications of both of these to space bounded complexity theory.

1 Introduction

While P vs NP is inarguably the most important problem that has remained unsolved in Computational Complexity, whenever we devise algorithms to solve almost any NP-complete problem on Graph Theory such as Graph Coloring or Vertex Cut or Travelling Salesman Problem and so on, we assume that the graph given to us as input is connected. Very often it suffices that our algorithm obtains the solution to the given problem for each connected component of the input graph, since these solutions are combined to obtain a solution to the given input instance of the graph theoretic problem. Due to this reason, the problem of deciding if given two vertices s and t in an input graph G, whether there exists a path that connects s and t in G is very fundamental and interesting. Let us call this problem as the st-connectivity problem (STCON).

The standard and well known polynomial time graph traversal algorithms of Breadth-First Search (BFS) and Depth-First Search (DFS) solve the STCON when the input graph G is directed or undirected. In the BFS algorithm, when a directed or an undirected graph is given as input, we systematically explore the input graph starting from s by visiting new vertices along paths of shortest length in a breadth-wise manner. Here we form a directed tree called the BFS tree with the vertex s as the root, and there exists a path between s and t in G if and only if the vertex t is in the BFS tree containing s as the root. In the case of the DFS algorithm, if the input graph G is directed, then we construct a maximal arborescence rooted at s such that t is in the same directed tree containing s if and only if there exists a directed path from s to t in G (an

[*]Email: `tcvijay@gmail.com`
[†]Email: `arvind@imsc.res.in`

arborescence in a directed graph G is a directed subgraph H of G such that every vertex has atmost one incoming edge and whose underlying undirected graph is a forest; an arborescence is maximal if it is not contained in any other arborescence of G). When the input graph G is undirected, the DFS algorithm solves the st-connectivity problem similar to the case when G is directed by considering each undirected edge as a pair of directed edges in opposite directions. Both these traversal algorithms have many applications to problems that can be solved in polynomial time. We refer those interested to [CLR22, Ski20, KV18] for more on these graph traversal algorithms.

1.1 Directed st-Connectivity

In Computational Complexity, the directed st-connectivity problem (DSTCON) is a well studied problem and it is known to be complete for the complexity class NL under logspace many-one reductions (see for example [AB09, Chapter 4]). Let us denote the complement of our problem DSTCON by $\overline{\text{DSTCON}}$ which is to decide if there does not exist any path from s to t in a directed graph G given as input. It is a well known result that DSTCON is logspace many-one reducible to its complement $\overline{\text{DSTCON}}$, which shows that NL = co-NL (for a nice algorithmic proof of this result, see [Sip13, Chapter 8]). This result generalizes and it is celebrated as the well known Immerman-Szelepcsényi Theorem which states that NSPACE($S(n)$)=co-NSPACE($S(n)$), where NSPACE($S(n)$) is any non-deterministic space bounded complexity class that contains NL and $S(n) \in \Omega(\log n)$. Restricted versions of the directed st-connectivity problem, such as for planar graphs, grid graphs and graphs embedded on surfaces have been investigated and many interesting results on their computational complexity using logarithmic space bounded counting classes are known [BTV09, ABC$^+$09].

1.2 Undirected st-Connectivity

Similar to DSTCON, the undirected st-connectivity problem (USTCON) is formally defined as follows: given an undirected graph $G = (V, E)$ and vertices $s, t \in V$, the USTCON problem is to determine if there exists an undirected path connecting s and t in G. It has been proved, by Omer Reingold in [Rei08], that the undirected st-connectivity problem (USTCON) is complete for the complexity class L under AC^0 many-one reductions. In this paper, we are interested in revisiting this important result [Rei08], and in explaining some key aspects of the $O(\log n)$ space algorithm of O. Reingold for USTCON.

1.2.1 Results prior to O. Reingold's deterministic $O(\log n)$ space algorithm for USTCON

O. Reingold in his paper [Rei08, Introduction] gives a nice account of the progress on the research work prior to obtaining a deterministic $O(\log n)$ space algorithm for USTCON starting from the randomized log-space algorithm due to Alelinuas *et.al.* [AKL$^+$79]. Following [AKL$^+$79], using pseudo-random generators, [AKS87, BNS89, NSW89, Nis92, INW94] obtain results aimed towards full derandomation of the complexity class RL, which would ultimately yield that USTCON is in L. Continuing this approach, in [ATW$^+$00] it is shown that USTCON$\in$ L$^{4/3}$ (where L^c is the complexity class of problems that can decided deterministically in space $\log^c(\cdot)$). This L$^{4/3}$ space upper bound for USTCON shown in [ATW$^+$00] is the most space-efficient algorithm for USTCON before the breakthrough result of O. Reingold in [Rei08], while V. Trifonov has independently presented an $O(\log n \log\log n)$-space deterministic algorithm for USTCON in [Tri05].

1.2.2 O. Reingold's approach for USTCON by improving connectivity of graphs using graph products

The $O(\log N)$-space deterministic algorithm designed by O. Reingold in [Rei08] decides if there exists an undirected path connecting vertices s and t in the input graph G by first improving the connectivity of vertices in G. Given an undirected graph G as input, we have a simple algorithm to determine if there exists an undirected path connecting s and t in G: enumerate all paths of *small* length starting from the vertex s, and find out if the vertex t is in any of the paths that we have enumerated. However this requires G to be well connected, since otherwise we will be unable to cover all the vertices in the connected component containing s using paths of *small* length. Moreover we need this algorithm to utilize only $O(\log N)$ space, where $N = |V(G)|$. To achieve this, O. Reingold uses the **zig-zag graph product** invented in [RVW02]. Using powering of graphs and the zig-zag graph product it is possible, by following the algorithm described in [Rei08], to transform the input graph G into an *expander graph* H which has good *expansion properties*, most notable of which is that the diameter of H is $O(\log N)$, where $N = |V(G)|$. As a result, by enumerating all paths of length $O(\log n)$ from the vertex s it is possible to determine if s and t are connected in G [Rei08]. It is a bit surprising to note that this entire algorithm of transforming G into an expander graph H and finally searching for the vertex t at a distance $O(\log N)$ from s is implementable using $O(\log N)$-space [Rei08], where $N = |V(G)|$.

2 Basic definitions and results

An undirected graph is called as a multi-graph if it contains one or more parallel edges or self-loops on vertices. In all the definitions and results we prove in this paper, we assume that we deal with undirected graphs that are multi-graphs unless otherwise mentioned.

2.1 Graph representations

We recollect the following facts about graph representations from [RVW02, Rei08]. There are several standard representations of graphs. For our results and purposes, we assume that the input graph G is given in terms of its adjacency matrix, which is the matrix whose (non-negative, integral) entry (u, v) equals the number of edges between u and v in G. Note that this representation allows graphs with self loops and parallel edges. As a result the graphs that we discuss may be simple or multi-graphs which means that vertices may have self-loops and parallel edges. A graph G is undirected if and only if its adjacency matrix is symmetric (implying that for every pair of distinct vertices $u, v \in V(G)$ there exists a directed edge from u to v if and only if there exists a directed edge from v to u).

A graph is D-regular if the sum of entries in each row and in each column is D (so exactly D edges are incident to every vertex). For every integer k, let us denote by $[k]$ the set $\{1, 2, ..., k\}$. Let G be a D-regular undirected graph on N vertices. A walk in a directed graph is a sequence of vertices $v_0 \sim v_1 \sim \cdots \sim v_r$ such that there exists a directed edge from v_i to v_{i+1}, where $1 \leq i \leq r$. A walk in a directed graph is a *closed walk* if its first and last vertices are the same. When considering a walk on an undirected D-regular graph G, we would like to assume that the edges, incident with each vertex of G are labeled from 1 to D in some arbitrary, but fixed, way. We can then talk about the i^{th} edge incident to a vertex v, and similarly about the i^{th} neighbor of v. A central insight of [RVW02] is that when taking a step on a graph from vertex v to vertex w, it may be useful to keep track of the edge traversed to get to w (rather than just remembering that we are now at w). This gives rise to a new representation of graphs through the following permutation on pairs of vertex name and edge label:

Definition 2.1. [Rei08, Definition 2.1] *For a D-regular undirected graph G, the rotation map $Rot_G : [N] \times [D] \to [N] \times [D]$ is defined as follows: $Rot_G(v, i) = (w, j)$ if the i^{th} edge incident to v leads to w, and this edge is the j^{th} edge incident to w.*

2.2 Graph products

Definition 2.2. [BM08, pp. 30] *The cartesian product of simple graphs G and H is the graph $G \square H$ whose vertex set is $V(G) \times V(H)$ and whose edge set is the set of all pairs $((u_1, v_1), (u_2, v_2))$ such that either $(u_1, u_2) \in E(G)$ and $v_1 = v_2$, or $(v_1, v_2) \in E(H)$ and $u_1 = u_2$.*

Thus for each edge (u_1, u_2) of G and (v_1, v_2) of H, there are four edges in $G \square H$, namely $((u_1, v_1), (u_2, v_1))$, $((u_1, v_2), (u_2, v_2)$, $((u_1, v_1), (u_1, v_2))$, and $((u_2, v_1), (u_2, v_2))$. Note that for any two simple graphs G and H, we have $G \square H = H \square G$.

Definition 2.3. [RVW02, Section 2.3, pp.168] *We define the tensor product of two graphs G_1 and G_2, denoted by $G_1 \otimes G_2$, by taking the tensor product of the adjacency matrices of G_1 and G_2. The set of vertices of $G_1 \otimes G_2$ is the cartesian product of the sets of vertices of G_1 and G_2, and (v_1, v_2) is adjacent to (u_1, u_2) in $G_1 \otimes G_2$ if and only if v_j is adjacent to u_j in G_j for $i = 1, 2$.*

Note that for any two simple graphs G and H, we have $G \otimes H = H \otimes G$.

Definition 2.4. [BM08, pp. 82] *The k^{th} power of a simple graph $G = (V, E)$ is the graph G^k whose vertex set is V, two distinct vertices being adjacent in G^k if and only if their distance in G is atmost k. The graph G^2 is referred to as the square of G, the graph G^3 as the cube of G.*

Let G be a undirected graph and let A be its adjacency matrix. Let us consider G as a directed graph G' in which every edge is replaced by a pair of directed edges in the opposite directions and every self-loop in G is replaced by a directed self-loop in G'. As a result, the number of directed closed walks of length k in G' is equal to the trace of A^k.

Now let G be an undirected graph, and without loss of generality assume that there exists a self-loop on every vertex of G. Let A be the adjacency matrix of G, then A^k is the adjacency matrix of G^k, for $k \in \mathbb{N}$. It is also easy to see that, for any square matrix A and $k \in \mathbb{Z}^+$, the eigenvalues of A^k are the k^{th} powers of the eigenvalues of A (see for example, [Str06, Section 5.2]).

While we need the above defined graph products to describe the $O(\log n)$-space deterministic algorithm in [Rei08], we refer the interested reader to [BM08, pp. 297, 364 & 316] for some more graph products such as the strong product, weak product and composition product (also known as the lexicographic product) of graphs.

3 Expander graphs

We now introduce expander graphs which are sparse graphs that are D-regular, for a constant $D > 0$ having linear number of edges, however have good connectivity properties in terms of *edge expansion* or *vertex expansion*. The spectrum of a graph G is the set of eigenvalues of the adjacency matrix A of G. We refer to [Big93, Chapters 2 and 3] for an easy introduction to combinatorial properties of the spectrum of ordinary and regular graphs. From these properties it follows that the largest eigenvalue of a D-regular undirected graph G is D, and G is connected if and only if the multiplicity of D in the spectrum of G is 1. In addition, $D = \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N \geq -D$, where λ_i is the i^{th} eigenvalue of the adjacency matrix A of a D-regular graph G, for $1 \leq i \leq N$. Also G is non-bipartite if and only if $\lambda_N > -D$.

3.1 Motivaton towards λ_2

In the eigenvalue spectrum of the adjacency matrix A of a D-regular graph G, the second largest eigenvalue (λ_2) of A is a key parameter that indicates the expansion properties of G. Now we motivate the importance of the second largest eigenvalue (λ_2) of the adjacency matrix A of a D-regular graph G.

For any $S \subseteq V(G)$, let $e(S)$ be the cardinality of the set of edges of the subgraph induced by S in G. Also let $e(S,\overline{S}) = |E(S,\overline{S})|$. A D-regular graph G is a δ-edge-expander (δ-expander for short) if for every set $S \subseteq V(G)$ of size at most $\frac{1}{2}|V(G)|$ there are at least $\delta D|S|$ edges connecting vertices in S and vertices in $\overline{S} = V(G) - S$, that is $e(S,\overline{S}) \geq \delta D|S|$.

Definition 3.1. *Let G be a D-regular graph on N vertices and let $S \subseteq V(G)$ such that $|S| \leq N/2$. Let us denote the set of edges connecting a vertex in S and a vertex in $\overline{S}$ by $E(S,\overline{S})$. We define the edge-expansion ratio of G as*

$$h(G) = \min_{\{S||S|\leq \frac{n}{2}\}} \frac{|E(S,\overline{S})|}{|S|}$$

Theorem 3.2. (Cheeger's inequality)[Che70] *Let G be a δ-expander and let A be the adjacency matrix of G with spectrum $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$. Then*

$$\frac{D-\lambda_2}{2} \leq h(G) \leq \sqrt{2D(D-\lambda_2)}.$$

The above theorem due to J. Cheeger [Che70] signifies the importance of $(D-\lambda_2)$ known as the *spectral gap* of G. In particular, a δ-edge-expander G has an expansion ratio $h(G)$ bounded away from zero if and only if its spectral gap $(D-\lambda_2)$ is bounded away from zero.

Proof.

Proof of left inequality: [ASS08, Theorem 1] Let A be the adjacency matrix of G and note that as A is symmetric we have $\lambda_2 = \max_{0\neq x\perp 1^n}\langle xA, x\rangle/\langle x, x\rangle$. For a set $S \subseteq V(G)$ let x_S be the vector satisfying $x_i = 1$ when $i \in S$ and $x_i = 0$ otherwise, and note that $\langle x_S A, x_S\rangle = 2e(S)$ and $\langle x_S A, x_S\rangle = e(S,\overline{S})$. Set $x = |\overline{S}| \cdot x_S - |S| \cdot x_{\overline{S}}$ and note that $x \perp 1^n$. Therefore,

$$\lambda_2(|S| + |\overline{S}|)|S||\overline{S}| = \lambda_2\langle x, x\rangle \geq \langle xA, x\rangle = 2|S|^2 e(\overline{S}) + 2|\overline{S}|^2 e(S) - 2|S||\overline{S}|e(S,\overline{S}). \tag{1}$$

As G is D-regular, we have $e(S) = \frac{1}{2}(D|S| - e(S,\overline{S}))$ and $e(\overline{S}) = \frac{1}{2}(D|\overline{S}| - e(S,\overline{S}))$. Plugging this into (1), solving for $e(S,\overline{S})$ and using $|S| \leq n/2$, we complete the proof by inferring that

$$e(S,\overline{S}) \geq (D-\lambda_2)|S||\overline{S}|/n \geq \frac{1}{2}(D-\lambda_2)|S|.$$

Proof of right inequality: [ASS08, Theorem 1] Let $Q = (DI - A)$ be Laplacian of G. Our goal is to prove that all but one of the eigenvalues of Q are at least $\frac{1}{2}\delta^2 D$. Let $z = (z_1, z_2, \ldots, z_n)$ be an eigenvector of Q with the smallest nontrivial eigenvalue λ, where $V(G) = \{1, 2, \ldots n\}$. Recall that for every set U of at most half the vertices of G there are at least $c|U|$ edges between U and its complement, where $c = \delta D$

is some positive constant. Without loss of generality assume that $m \le n/2$ of the entries of z are positive (otherwise, replace z by $-z$), and that $z_1 \ge z_2 \ge \cdots \ge z_m \ge 0 \ge z_{m+1} \ge \cdots \ge z_n$. Define $x_i = z_i$ for $i \le m$, and $x_i = 0$ otherwise. Since $x_j = 0$ for all $j > n/2$,

$$\sum_{ij\in E(G)} |x_i^2 - x_j^2| = \sum_{ij\in E, i<j} (x_i^2 - x_j^2) \ge \sum_{i:i\le n/2} (x_i^2 - x_{i+1}^2)ci = c\sum_{i=1}^{n} x_i^2. \tag{2}$$

Note that $(Qz)_i = \lambda z_i$ for all i and hence $\lambda = \frac{\sum_{i=1}^m (Qz)_i z_i}{\sum_{i=1}^m z_i^2}$. However,

$$\sum_{i=1}^{m}(Qz)_i z_i = \sum_{i=1}^{m}(dz_i^2 - \sum_{j,ij\in E} z_i z_j) = \sum_{i,j\le m, ij\in E} (z_i - z_j)^2 + \sum_{i\le m, j>m, ij\in E} z_i(z_i - z_j) \ge \sum_{ij\in E}(x_i - x_j)^2.$$

As $\sum_{i=1}^m z_i^2 = \sum_{i=1}^n x_i^2$ we conclude, using Cauchy-Schwartz inequality (twice) that

$$\lambda \ge \frac{\sum_{ij\in E}(x_i - x_j)^2}{\sum_{i=1}^n x_i^2} = \frac{\sum_{ij\in E}(x_i - x_j)^2 \sum_{ij\in E}(x_i + x_j)^2}{\sum_i x_i^2 \sum_{ij\in E}(x_i + x_j)^2} \ge \frac{(\sum_{ij\in E} |x_i^2 - x_j^2|)^2}{\sum_i x_i^2 2D \sum_i x_i^2} \ge \frac{c^2}{2D},$$

where the last inequality follows from (2). Therefore, $\lambda \ge \frac{c^2}{2D} = \frac{1}{2}\delta^2 D$. ■

3.2 Existence of Expander graphs

The existence of expander graphs was first proved by M. S. Pinsker [Pin73],[AS16, pp.152]. As stated in [RVW02, pp. 158], standard probabilistic arguments ([Pin73]) show that almost every constant (≥ 3) degree graph is an expander (also see [ASS08, Theorem 2]). We recall (as stated in [JM87, pp. 343]) that the first explicit constructions of expanders are due to G. A. Margulis and also due to Gabber and Galil [GG81]. An elementary example of undirected 3-regular expander graphs on p vertices, where p is a prime, is a Cayley graph defined as follows: let $G = (\mathbb{Z}_p, E)$, where for any vertex $x \in \mathbb{Z}_p$ edges $(x, x+1), (x, x-1)$ and (x, x^{-1}) are in E. Here we assume that the multiplicative inverse of x is x^{-1} and that the multiplicative inverse of 0 is itself.

Definition 3.3. *Let $G = (V, E)$ be a simple undirected graph and let $N = |V|$. For any $S \subseteq V$, we denote the set of neighbors of vertices in S by $\Gamma(S)$. We say that G is an $(\alpha, D-2)$-expander if for every $S \subseteq V$ such that $|S| \le \alpha N$ we have $\Gamma(S) \ge (D-2)|S|$.*

Theorem 3.4. *Let $\mathcal{G}_{N,D}$ be the set of simple undirected bipartite graphs with bipartite sets L, R of size N each, and which is left regular having degree D for every vertex in L. Then for any D, there exists an $\alpha(D) > 0$ such that for all N*

$$\Pr[G \textit{ is an } (\alpha N, D-2)\textit{-expander}] \ge 1/2,$$

where G is chosen uniformly at random from $\mathcal{G}_{N,D}$.

Proof. To choose G in $\mathcal{G}_{N,D}$ uniformly at random, we choose D neighbors for each vertex in L uniformly at random. For $K \le \alpha N$, let us define

$$p_K = \Pr[\exists S \subseteq L : |S| = k, |\Gamma(S)| < (D-2)|S|].$$

So to prove that Pr[G is not an $(\alpha N, D-2)$ − expander] $\geq 1/2$, it suffices to show that $\sum_K p_K \geq \frac{1}{2}$. Now assume that there is a set S of size K and $|\Gamma(S)| < (D-2)|S|$. Since the total number of neighbors of vertices in S, counted with multiplicity, is DK, there are at least $2K$ repeats among all the neighbors of vertices in S. We calculate the quantity,

$$\Pr[\text{at least 2K repeats among all the neighbors of vertices in S}] \leq \binom{KD}{2K}\left(\frac{KD}{N}\right)^{2K}.$$

Here the binomial coefficient represents the number of ways of choosing $2K$ neighbors to be repeats from the (multi-)set of all neighbors of vertices in L, and the fraction $\frac{KD}{N}$ represents an upper bound on the probability that any given choice of a neighbor is a repeat. Since there are $\binom{N}{K}$ possibilities for S, we have,

$$\begin{aligned} p_K &\leq \binom{N}{K}\binom{KD}{2K}\left(\frac{KD}{N}\right)^{2K} \\ &\leq \left(\tfrac{N\mathrm{e}}{K}\right)^K \cdot \left(\tfrac{KD\mathrm{e}}{2K}\right)^{2K} \cdot \left(\tfrac{KD}{N}\right)^{2K} \\ &\leq \left(\tfrac{cKD^4}{N}\right)^K, \end{aligned}$$

where $c = \mathrm{e}^3$. By setting $\alpha = 1/(cD^4)$ and $K \leq \alpha N$, we know that $p_K \leq 4^{-K}$ and

$$\Pr[G \text{ is not an } (\alpha N, D-2)\text{−expander}] \leq \sum_{K=1}^{\alpha N} p_K \leq \sum_{K=1}^{\alpha N} 4^{-K} \leq 1/2.$$

■

Corollary 3.5. *Let G be a simple undirected graph chosen at random from the set of all simple D-regular undirected graphs on N vertices. Then, for any D, there exists an $\alpha(D) > 0$ such that for all N,*

$$\Pr[G \textit{ is an } (\alpha N, D-2)\textit{-expander}] \geq 1/2.$$

Proof. Given an D-regular undirected graph G on $\{1, \ldots N\}$ vertices, let us consider the D-regular undirected bipartite graph H on $2N$ vertices, labeled by $\{1, \ldots, 2N\}$, obtained by having a partition of the vertex set of H into $L = \{1, \ldots N\}$ and $R = \{N+1, \ldots, 2N\}$ containing N vertices each such that for every undirected edge (i, j) in G there exists an undirected edge $(i, j+N)$ in H. It is then easy to see that G is an $(\alpha(D), D-2)$-expander if and only if H is also an $(\alpha(D), D-2)$-expander. Our result now follows from Theorem 3.4. ■

Several explicit constructions of expanders which are Cayley graphs are known. Constructions of families of bipartite expanders which are Cayley graphs having constant degree 5, 7, 8 shown in [GG81, JM87] are some of the earliest examples of expanders. As stated in [AS16, pp.153]: It is known (see [Nil91]) that the second largest eigenvalue of any D-regular graph with diameter k is at least $2\sqrt{D-1}(1-O(1/k))$. Therefore, in any infinite family of D-regular graphs, the limsup of of the second largest eigenvalue is at least $2\sqrt{D-1}$ (also see [LPS88, Proposition 4.2]). The importance of the number $2\sqrt{D-1}$ is due to N. Alon and R. Boppana [Alo86].

Definition 3.6. [LPS88, Definition 1.1] *Let $X_{N,D}$ be a D-regular graph on N vertices. A graph $X_{N,D}$ is a* **Ramanujan graph** *if $\lambda_2(X_{N,D}) \leq 2\sqrt{D-1}$.*

[LPS88, pp.262] have given explicit constructions of infinite families of D-regular Cayley graphs G_i, with the second largest eigenvalue $\lambda_2(G_i) \leq 2\sqrt{D-1}$ and hence these graphs are in fact Ramanujan graphs.

3.3 Spectral expansion and Vertex expansion

The normalized adjacency matrix M of a D-regular graph G is the matrix obtained by dividing the adjacency matrix A of G by D. Let G be a D-regular graph and let M be the normalized adjacency matrix of G. The largest eigenvalue of M is $\lambda_1 = 1$ with eigenvector $u = (1/N, \ldots, 1/N)^T$. Then the second largest eigenvalue is given by,

$$\lambda_2 = \max_{\|x\|=1, x \perp u} \|Mx\|.$$

If π is a probability distribution on the vertices of G (represented as a vector), then we can write $\pi = u + \pi^\perp$, where $\pi^\perp \perp u$. Since G is D-regular and assuming π as the initial distribution,

$$M\pi - u = M(u + \pi^\perp) - u = Mu - u + M\pi^\perp = M\pi^\perp.$$

Thus,

$$\|M\pi - u\|^2 = \|M\pi^\perp\|^2 \leq \lambda_2^2 \|\pi^\perp\|^2 = \lambda_2^2 \|\pi - u\|^2.$$

Definition 3.7. *A graph G has spectral expansion λ if $\lambda_2(G) \leq \lambda$.*

Definition 3.8. *Given a probability distribution π, the collision probability of π is* $\mathrm{Coll}(\pi) = \|\pi\|^2 = \sum_x \pi_x^2$.

Lemma 3.9. $\mathrm{Coll}(\pi) = \|\pi - u\|^2 + 1/N$.

Proof. Write $\pi = u + \pi^\perp$. Then,

$$\|\pi\|^2 = \|u\|^2 + \|\pi^\perp\}^2 = 1/N + \|\pi - u\|^2.$$

■

Note that $M\pi$ is also a probability distribution and using the lemma, we can compute the associated collision probability:

$$\mathrm{Coll(M\pi)} - 1/N = \|M\pi - u\|^2 \leq \lambda^2 \|\pi - u\|^2 = \lambda^2(\mathrm{Coll}(\pi) - 1/N).$$

Given a probability distribution π, let the *support* of x be $\mathrm{support}(\pi) = \{x | \pi_x \neq 0\}$.

Lemma 3.10. *Let π be a probability distribution. Then* $\mathrm{Coll}(\pi) \geq 1/|\mathrm{support}(\pi)$.

Proof. Let $m = |\mathrm{support}(\pi)|$. We claim that if $x_1 + x_2 + \cdots x_m = x$, then $x_1^2 + x_2^2 + \cdots x_m^2$ is minimized (with value x/m) when $x_1 = x_2 = \cdots x_m = x/m$. This easily follows from the fact that $x^2 + y^2 \geq ((x+y)/2)^2 + ((x+y)/2)^2$. Thus $\mathrm{Coll}(\pi) \geq 1/m$ and we are done. ■

Theorem 3.11. *If G has spectral expansion λ, then for all $\alpha > 1$, G has vertex expansion $(\alpha N, \frac{1}{(1-\alpha)\lambda^2 + \alpha})$.*

Proof. Let $|S| \leq \alpha N$. Choose π to be the probability distribution that is uniform on S and 0 on the complement of S. Then,

$$\mathrm{Coll}(\pi) = 1/|S| \qquad \text{and} \qquad \mathrm{Coll}(M\pi) \geq 1/|\mathrm{support}(M\pi)| = 1/|\Gamma(S)|.$$

Also,

$$1/|\Gamma(S)| - 1/N \leq \lambda^2(1/|S| - 1/N).$$

However $N \geq |S|/\alpha$, so solving the above inequality gives

$$|\Gamma(S)| \geq \frac{|S|}{(1-\alpha)\lambda^2 + \alpha}.$$

Thus G is an $(\alpha N, 1/((1-\alpha)\lambda^2 + \alpha))$-expander. ■

Theorem 3.12. *Let G be a D-regular graph on N vertices which is a $(N/2, 1+\alpha)$-expander and let $\lambda_2(G)$ be the second largest eigenvalue of the normalized adjacency matrix M of G. If M has all non-negative eigenvalues, then G is a λ-spectral expander for $\lambda = 1 - \alpha^2/(d(8+4\alpha^2))$.*

Proof. Let x be an eigenvector with eigenvalue $\lambda_2 M)$. Since $x \perp u$, the vector x has both positive and negative entries. Let $V_+ = \{i : x_i > 0\}$ and $V_- = \{i : x_i \leq 0\}$. Without loss of generality $|V_i| \leq N/2$. Let $\overline{x}$ be the vector that agrees with x on V_+ and is 0 elsewhere.

Note that $\langle \overline{x}, \overline{x} \rangle = \langle x, \overline{x} \rangle$, so it can be shown that

$$\lambda_2(M) = \frac{\lambda_2 \langle x, \overline{x} \rangle}{\langle x, \overline{x} \rangle} = \frac{\lambda_2 \langle x, \overline{x} \rangle}{\langle \overline{x}, \overline{x} \rangle} = \frac{\langle Ax, \overline{x} \rangle}{\langle \overline{x}, \overline{x} \rangle}.$$

Also,

$$\begin{aligned}
\lambda_2(M)\langle \overline{x}, \overline{x} \rangle &= \langle Ax, \overline{x} \rangle \\
&= \sum_{i,j} A_{ij} x_j \overline{x}_i \\
&= \|x\|^2 - \frac{1}{d}\left(D\|x\|^2 - \sum_{i \in V_+, \{i,j\} \in E} \overline{x}_i x_j \right) \\
&= \|x\|^2 - \frac{1}{d}\left(D\|x\|^2 - \sum_{i \in V_+, \{i,j\} \in E} \overline{x}_i \overline{x}_j - \sum_{i \in V_+, j \in V_-, \{i,j\} \in E} \overline{x}_i \overline{x}_j \right) \\
&= \|x\|^2 - \frac{1}{d}\left(D\|x\|^2 - \sum_{\{i,j\} \in E} \overline{x}_i \overline{x}_j \right) \\
&= \|x\|^2 - \frac{1}{D} \sum_{\{i,j\} \in E} (\overline{x}_i - \overline{x}_j)^2
\end{aligned}$$

$$\lambda_2 \leq 1 - \frac{\sum_{\{i,j \in E\}} (\overline{x}_i - \overline{x}_j)^2}{D \sum_{i \in V} \overline{x}_i^2}. \tag{3}$$

Build a new (weighted directed) graph H as follows. Let

$$V(H) = \{s\} \cup \{v_i : i \in V_+\} \cup \{w_j : j \in V\} \cup \{t\}.$$

For all $i \in V_+$, put the arcs (s, v_i) in H with capacity $1 + \alpha$. For each $i \in V_+$ and $j \in V$, where j is a neighbor of i in G, put the arcs (v_i, w_j) in H with capacity 1. Finally, for each $j \in V$, put the arcs (w_j, t) in H with capacity 1.

We claim that the minimum cut in this graph is $(1 + \alpha)|V_+|$. A cut of this size is given by the set of arcs $\{(s, v_i) : i \in V_+\}$. Given any other cut C, let $W = \{i \in V_+ : (s, v_i) \notin C\}$. For each $j \in \Gamma(W)$, there must be an arc in C adjacent to w_j. But $\Gamma(W) \geq (1 + \alpha)|W|$, so the capacity of C must be at least $(1 + \alpha)|V_+ - W| + |\Gamma(W)| \geq (1 + \alpha)|V_+|$ (since $|W| \leq N/2$). So the minimum cut has capacity $(1 + \alpha)|V_+|$.

By the Max-flow min-cut Theorem [CLR22, Section 24.2, Theorem 24.6], there exists a flow on H of size $(1 + \alpha)|V_+|$. In particular, note that the flow through each vertex v_i must be $1 + \alpha$. Reading off the flow along arcs (v_i, w_j), it follows that there is a function $F : V \times V \to \mathbb{R}$ satisfying the following conditions (here $\tilde{E}$ denotes the set of ordered pairs (i, j) where $\{i, j\} \in E$, so that each edge in E is counted twice in $\tilde{E}$):

1. $0 \leq F(i, j) \leq 1$ for all $i, j \in V$.
2. $F(i, j) = 0$ if $i \notin V_+$ or $(i, j) \notin \tilde{E}$.
3. $\sum_{j:(i,j)\in\tilde{E}} F(i, j) = 1 + \alpha$ for each $i \in V_+$.
4. $\sum_{i:(i,j)\in\tilde{E}} F(i, j) \leq 1$ for each $j \in V$.

We need to calculate two bounds involving F in order to bound $\lambda_2(G)$. Keeping in mind that $2(a^2 + b^2) \geq (a + b)^2$ for all real numbers a and b, we find:

$$\begin{aligned}
\sum_{(i,j\in\tilde{E}} F^2(i,j)(\overline{x}_i + \overline{x}_j)^2 &\leq 2 \sum_{(i,j)\in\tilde{E}} F^2(i,j)(\overline{x}_i^2 + \overline{x}_j^2) \\
&= 2 \sum_{(i,j)\in\tilde{E}} \overline{x}_i^2 \left(\sum_{(i,j)\in\tilde{E}} F^2(i,j) + \sum_{(i,j)\in\tilde{E}} F^2(j,i) \right) \\
&\leq (4 + 2\alpha^2) \sum_{i\in V} \overline{x}_i^2. \\
\sum_{(i,j)\in\tilde{E}} F(i,j)(\overline{x}_i^2 - \overline{x}_j^2) &= \sum_{i\in V} \overline{x}_i^2 \left(\sum_{(i,j)\in E} F(i,j) - \sum_{(i,j)\in\tilde{E}} F(j,i) \right) \\
&\geq \alpha \sum_{i\in V} \overline{x}_i^2.
\end{aligned}$$

Note that in the third line, we used the fact that if $x_1 + \cdots + x_n = 1 + \alpha$ and $0 \leq x_i \leq 1$ for all i, then $x_1^2 + \cdots + x_n^2 \leq 1 + \alpha^2$. Multiplying equation (3) by

$$1 = \frac{\sum_{(i,j)\in\tilde{E}} F^2(i,j)(\overline{x}_i + \overline{x}_j)^2}{\sum_{(i,j)\in\tilde{E}} F^2(i,j)(\overline{x}_i + \overline{x}_j)^2}$$

and using Cauchy-Schwartz inequality, we get

$$\begin{aligned}
\lambda_2(G^2) &\leq 1 - \frac{\sum_{(i,j)\in E}(\overline{x}_i - \overline{x}_j)^2}{D\sum_{i\in V}\overline{x}_i^2} \\
&= 1 - \frac{\sum_{(i,j)\in E}(\overline{x}_i - \overline{x}_j)^2 \cdot \sum_{(i,j)\in\tilde{E}} F^2(i,j)(\overline{x}_i + \overline{x}_j)^2}{D\sum_{i\in V}\overline{x}_i^2 \cdot \sum_{(i,j)\in\tilde{E}} F^2(i,j)(\overline{x}_i + \overline{x}_j)^2} \\
&\leq 1 - \frac{\left(\sum_{(i,j)\in\tilde{E}} F(i,j)(\overline{x}_i^2 - \overline{x}_j^2)\right)^2}{2D(4+2\alpha^2)\cdot\left(\sum_{i\in V}\overline{x}_i^2\right)^2} \\
&\leq 1 - \frac{\alpha^2}{D(8+4\alpha^2)}.
\end{aligned}$$

This completes the proof. ■

We now move to the general case, when the eigenvalues of M need not all be non-negative.

Corollary 3.13. *Let G be a D-regular graph on N vertices which is a $(N/2, 1+\alpha)$-expander. Then G is also a λ-spectral expander for $\lambda = \sqrt{1 - \alpha^2/(D^2(8+4\alpha^2))}$.*

Proof. Consider the graph G^2 (recall Definition 2.4). If the normalized adjacency matrix of G is M, then the normalized adjacency matrix of G^2 is M^2. Also G^2 is D^2-regular and has all non-negative eigenvalues. Finally, G^2 is a $(N/2, 1+\alpha)$-expander, as follows: If S is a subset of the vertices of size at most $N/2$, then $|\Gamma(S)| \geq (1+\alpha)|S|$. Choose a subset S' of $\Gamma(S)$ with $|S| \leq |S'| \leq N/2$. Then $|\Gamma(\Gamma(S))| \geq |\Gamma(S')| \geq (1+\alpha)|S'| \geq (1+\alpha)|S|$. But $\Gamma(\Gamma(S))$ is the neighborhood of S in G^2, and S was an arbitrary subset of vertices of size at most $N/2$, so G^2 is a $(N/2, 1+\alpha)$-expander.

By Theorem 3.12, $\lambda_2(G^2) < 1 - \alpha^2/(D^2(8+4\alpha^2))$. Since the eigenvalues of G^2 are the squares of the eigenvalues of G, taking the square-root of the right hand side proves the corollary. ■

3.4 Significance of λ_2

Definition 3.14. *Let G be a D-regular graph on N vertices and let A be the adjacency matrix of G. We say that G is a (N, D, λ)-expander if the second largest eigenvalue of A (denoted by λ_2) is atmost λ.*

The following theorem, known as the Expander Mixing Lemma, shows that every (N, D, λ)-expander graph is close to acting as a random D-regular graph on N vertices.

Theorem 3.15. [AC88] *Let $G = (V, E)$ be a D-regular, N-vertex graph with spectral expansion λ. Then $\forall S, T \subseteq V$, we have*

$$\left| |E(S,T)| - \frac{D|S|\cdot|T|}{N} \right| \leq \lambda\sqrt{|S||T|},$$

where $E(S,T)$ is the set of edges with one vertex in S and the other vertex in T.

Note that in the above statement, $|E(S,T)|$ is the number of edges with one vertex in S and the other vertex in T. Also in a random D-regular graph G, the expected number of edges between S and T in G is $\frac{D|S|\cdot|T|}{N}$, since the Pr[choosing a vertex randomly from any $S \subseteq V$] $= \frac{|S|}{N}$ and $|E| \leq ND$. As a result smaller λ implies that the expander graph G is more random.

Proof. Let A be the adjacency matrix of G. Since G is undirected, A is real and symmetric. As a result its eigenvalues are real and there exists a set of orthonormal eigenvectors $\{v_1, \dots v_N\}$ of A which is a basis of $\mathbb{R}^N$. Let $\mathbf{1}_S$ and $\mathbf{1}_T$ be the characteristic vectors S and T. It is easy to see that

$$|E(S,T)| = \sum_{u \in S, v \in T} A_{uv} = \mathbf{1}_S^T A \mathbf{1}_T.$$

Expressing $\mathbf{1}_S$ and $\mathbf{1}_T$ as a linear combination of the orthonormal eigenvectors basis of $\mathbb{R}^N$, we have

$$\begin{aligned} |E(S,T)| &= (\sum_i \alpha_i v_i)^T A (\sum_j \beta_j v_j) \\ &= \sum_i \sum_j \beta_j \alpha_i v_i^T \lambda_j v_j \\ &= \sum_i \alpha_i \beta_i \lambda_i \\ &= \alpha_1 \beta_1 \lambda_1 + \sum_{i=2}^N \alpha_i \beta_i \lambda_i. \end{aligned}$$

We know that

$$\alpha_1 = \frac{|S|}{\sqrt{N}}, \beta_1 = \frac{|T|}{\sqrt{N}}, \lambda_1 = D.$$

Therefore,

$$\alpha_1 \beta_1 \lambda_1 = \frac{D}{N}|S||T|$$

Then we get,

$$\begin{aligned} |E(S,T)| - \frac{D}{N}|S||T| &= \sum_{i=2}^N \alpha_i \beta_i \lambda_i \\ \left| |E(S,T)| - \frac{D}{N}|S||T| \right| &= \left| \sum_{i=2}^N \alpha_i \beta_i \lambda_i \right| \end{aligned}$$

We can bound the right side as,

$$\begin{aligned} \left| \sum_{i=2}^N \alpha_i \beta_i \lambda_i \right| &\leq \lambda \cdot \sum_{i=2}^N |\alpha_i \beta_i| \\ &\leq \lambda \left(\sum_{i=2}^N |\alpha_i|^2 \right)^{\frac{1}{2}} \left(\sum_{i=2}^N |\beta_i|^2 \right)^{\frac{1}{2}} \\ &\leq \lambda \cdot \|\mathbf{1}_S\|_2 \|\mathbf{1}_T\|_2 \\ &= \lambda \sqrt{|S||T|} \end{aligned}$$

As a result,

$$\left| |E(S,T)| - \frac{D|S| \cdot |T|}{N} \right| = \left| \sum_{i=2}^N \alpha_i \beta_i \lambda_i \right|$$

which implies,

$$\left| |E(S,T)| - \frac{D|S| \cdot |T|}{N} \right| \leq \lambda \sqrt{|S||T|}.$$

$\blacksquare$

A result proved in [BL06, Lemma 3.3] is close to being the converse of the above shown Expander Mixing Lemma.

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